SMARTPHONE MOTION SENSORS AND HEALTHCARE: On The Cusp Of A Disruption?

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On a balmy mid-April afternoon, amidst the cacophony of passionate deliberations ringing the walls of Paul Brest Hall on Stanford University's sylvan green campus, a picture began to emerge. It was the annual mHealth Connect conference. Most of the talks by academic stalwarts, healthcare entrepreneurs and venture capitalists were harping on a common thesis, "The growing nexus between mobile sensors and artificial intelligence (AI) in the healthcare domain."

Amidst ardent calls for Silicon Valley-styled disruption of the ailing healthcare sector, it became ostensibly clear that three facets of personal healthcare were going to see important changes, driven by the emergence of sensors-laden smartphones as the centerpiece of democratisation of healthcare hardware. In the next few paragraphs you will be exposed to the following three domains earmarked to be the recipients of the disruption: gait- and posture-related disorders, geriatric care and treatment of neurodegenerative disorders (Parkinson's disease, multiple sclerosis, Alzheimer's disease and Huntington disease).

At first glance it might seem as if ushering these advancements demands dramatic popularisation of highly-specialised and potentially expensive medical sensors, such as photo-plethysmo-gram (PPG) heart rate monitor sensor found thus far only in high-end phones such as Samsung S8. Far from it, we will see that many solutions catering to a wide array of challenges in all three above-mentioned domains, will, in fact, be fuelled by the data emanating from the humble triad of inertial measurement unit sensors, or motion sensors, that is, accelerometers, gyroscopes and magnetometers.

Three facets of personal healthcare—gait- and posture-related disorders, geriatric care and treatment of neurodegenerative disorders—are set to see important changes, driven by the emergence of sensors-laden smartphones as the centerpiece of democratisation of healthcare hardware

With this background, let us dive deeper into the specific domains and the success stories that have emerged.

Gait- and posture-related disorders

Paleoanthropologists (those who study fossil hominids) believe that approximately four million years ago, the now-extinct hominin *Australopithecus afarensis* finally mastered the art of performing a series of complicated, cooperative and coordinated manipulation of the musculoskeletal and locomotor controller systems, resulting in a physical activity that we now take for granted—the upright human bipedal walk!

The repetitive pattern spanning two successive steps or one complete stride constitutes what is formally studied as a gait cycle. A gait cycle begins when the heel of one foot strikes the ground and culminates the instant when the same foot makes contact with the ground again.

As per Rancho Los Amigos system, a single gait cycle consists of eight temporal phases, which include initial contact, loading response, mid-stance, terminal stance, pre-swing, initial swing, mid-swing and late-swing phases.

Fine grained analysis of gait reveals an astonishing array of clues about a person's state of general health, both from the orthopedic as well as the neuromuscular perspective.

From an orthopedic viewpoint, study of gait kinematic deviations from the normative model is an integral part of diagnosis of conditions such as developmental dysplasia of the hip (DDH), hip fracture and chronic ankle instability.

Stanford Motion and Gait-Analysis Laboratory lists eight basic pathological gaits that can be attributed to neurological conditions. These are hemiplegic, spastic diplegic, neuropathic, myopathic, Parkinsonian, choreiform, ataxic (cerebellar) and sensory.

A typical gait analysis study entails a physical exam, followed by video analysis and possibly electromyographic (EMG) measurements of muscle activity during movement. Enter the scene, motion sensors!

Modern-day smartphone sensors, such as accelerometers and gyroscopes, allow sampling rates of as high as 400 samples/sec. These facilitate highly precise measurements of the gait cycle that can be done by the patient just by placing the phone in his/